

Master's Comprehensive Examination

Part II

Applied (90 minutes)

December 4, 1999

Instructions: Answer three out of four questions. Open book and open notes.

1. High levels of blood fat contribute to atherosclerosis (hardening of the arteries), which is an underlying cause of heart attacks and stroke. Dr. Robert Knopp of the University of Washington School of Medicine in Seattle analyzed 381 diabetics and found that women had higher levels of LDL, or "bad cholesterol," than men and lower levels of HDL, or "good cholesterol." From the following simulated data, determine whether the level of LDL in women is greater than that in men:

Women: $n_1 = 201$, $\bar{y}_1 = 52.35$, $s_1 = 6.55$
 Men: $n_2 = 180$, $\bar{y}_2 = 47.28$, $s_2 = 17.29$

2. These summary data resulted from an experiment involving a treatment with four levels:

	Treatment				Source	SS	df
	A1	A2	A3	A4			
n	12	12	12	12	Treatment	756.5	3
$\sum y_i$	135	152	147	184	Error	261.3	44

Analyze the means using Tukey's multiple comparison procedure with $\alpha = .01$.

3. Three factors are studied for their effect on the horsepower necessary to remove a cubic inch of metal per minute. The factors are feed rate at two levels, tool condition at two levels, and tool type at two levels. For each treatment combination three observations are taken and all 24 observations of horsepower are made in a completely randomized order. The data appear below.

Tool Type	Feed Rate			
	0.011		0.015	
	Tool Condition		Tool Condition	
	<i>Same</i>	<i>New</i>	<i>Same</i>	<i>New</i>
Utility	0.576	0.548	0.514	0.498
	0.576	0.555	0.515	0.519
	0.565	0.540	0.518	0.504
Precision	0.526	0.547	0.494	0.521
	0.542	0.524	0.504	0.480
	0.548	0.525	0.530	0.494

Do a complete ANOVA on these data as a 2^f experiment and state your conclusions.